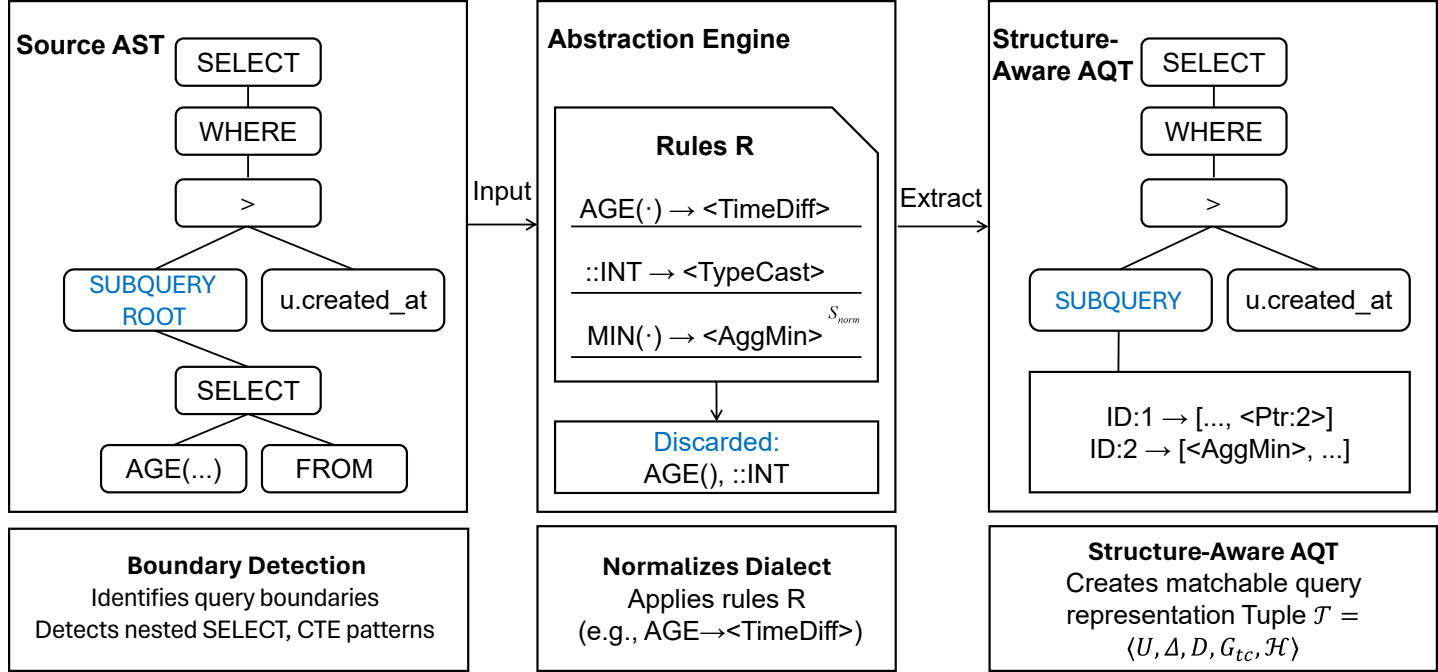

Source AST
SELECT
WHERE
>
SUBQUERY ROOT
u.created_at
SELECT
AGE(...)
FROM
Input
Abstraction Engine
Rules R
AGE(·) → <TimeDiff>
::INT → <TypeCast>
MIN(·) → <AggMin>
S_{norm}
Discarded:
AGE(), ::INT
Extract
Structure-Aware AQT
SELECT
WHERE
>
SUBQUERY
u.created_at
ID:1 → [..., <Ptr:2>]
ID:2 → [<AggMin>, ...]
Boundary Detection
Identifies query boundaries
Detects nested SELECT, CTE patterns
Normalizes Dialect
Applies rules R
(e.g., AGE→<TimeDiff>)
Structure-Aware AQT
Creates matchable query representation Tuple $\mathcal{T} = \langle U, \Delta, D, G_{tc}, \mathcal{H} \rangle$